

Smart City Concern: An AI-Driven Multimodal Civic Grievance Management System

Mr. R. Praveen Kumar¹, P. Sam Benny², Ch. Manideep³,
G. Sushmitha⁴, Md. Athfaan⁵

¹Assistant Professor, ^{2,3,4,5}UG Scholar

Department of Computer Science and Engineering (AI & ML)

Vignan Institute of Technology and Science, Nalgonda, Telangana-508284

E-mail: praveenvitshyd@gmail.com

, sambenny0005@gmail.com, ch.manideep2004@gmail.com,

sushmitha.guguloth2004@gmail.com, athfaan05@gmail.com.

Abstract—Manual Urban civic grievance systems are usually based on manual processing and resulting to delayed routing, misclassification and lack of transparency. A smart city concierge system is a multimodal AI-driven civic grievance intelligence that is proposed in this paper. It combines web-based complaints, social media monitoring via X (formerly Twitter). The system leverages transformer-based zero-shot classification for automated department routing, Named Entity Recognition for structured information extraction, sentiment-driven priority scoring, and OCR-based image text extraction for multimodal analysis. Duplicate complaints are detected using TF-IDF and cosine similarity to reduce redundancy. A scalable layered architecture using Flask and MongoDB ensures efficient ticket management and analytics. Experimental evaluation demonstrates improved classification accuracy, reduced manual workload, and enhanced response efficiency, making the proposed research suitable for practical smart city governance.

Index Terms—Smart City; Civic Grievance Management; Multimodal Analysis; Natural Language Processing; Transformer Models; Zero-Shot Classification; Social Media Mining; Image-Based Analysis; Optical Character Recognition (OCR); Sentiment Analysis; Duplicate Detection; MongoDB; X (formerly Twitter) API Integration; Urban Governance Automation.

I. INTRODUCTION

Fast urbanization has led to the rise of many civic complaints, which are addressed by the municipal authorities using the web portals, call centers and social media. Nonetheless, most of the existing grievance redressal systems remain manual making them slow in categorizing complaints, directing issues in the wrong direction, duplicating complaints and lack of transparency. These inefficiencies decrease the efficiency of response and have negative impacts on the satisfaction of citizens.

Citizens have also started reporting civic concerns online, through tagging municipal authorities with the increasing use of social media sites like X (formerly Twitter) (X). It is hard to process such complaints manually since they can be different in terms of language, format, and multimedia content. Images that offer a lot of contextual information are also often used in many complaints and cannot be effectively analyzed through traditional text-based systems.

To resolve them, the given research suggests Smart City Concern: A Multimodal AI-based Civic Grievance Intelligence

System. Natural Language Processing (NLP), transformer-based zero-shot classification, Named Entity Recognition (NER), sentiment-based priority scoring, and Optical Character Recognition (OCR) are used to analyze both text and pictures in the system. It classifies, detects duplicates, routes complaints to a department, and assigns priorities with a scalable architecture implemented on Flask and MongoDB, enhancing efficiency and minimizing manual tasks, as well as helping the city to be smarter and smarter.

II. RELATED WORKS

The fast development of Artificial Intelligence has opened up new opportunities in bettering the city administration and provision of civic services. The analytics systems developed based on AI can help authorities to comprehend the feedback of citizens and govern the resources of the population more effectively. Raj et al. designed a predictive analytics predictive AI framework of real-time civic engagement and decision-making in smart cities [1].

Likewise, Lifelo et al. addressed the issue of using AI technologies to create smart city infrastructure and digital services in the context of sustainability [6].

Varde et al. also focused on the significance of the AI basis and implementation in various fields of a smart city such as governance and community safety [7].

Nikhra et al. emphasized the models of AI-driven participatory governance to motivate citizens to report and actively monitor urban problems and promote urban management [9].

The concept of digital grievance redressal systems has been actively researched to enhance the work of the citizen-government interaction and the simplification of complaint management. Mande et al. proposed the use of Civic Connect, a web-based tool enabling citizens to report any issues of interest to the community and monitor their status of resolution [5].

Kotwal et al. suggested the creation of a web-based grievance portal that was aimed at enhancing transparency and accessibility in the complaint submission and tracking [11].

Verma et al. analyzed the digital grievance redressal processes to help clean and smarter urban governance [10].

Another concept introduced by Kumari was CIVIC EYE, a

machine-assisted computerized intelligence tool capable of expanding the level of civic engagement and enhancing the rate of public safety monitoring [14].

As the use of social media expands rapidly, researchers have also sought to use social media as a useful tool of civic feedback and opinion. In their article, Khan et al. summarized the social media informatics and spoke about the use of AI methods to analyze citizen-generated information to manage the city on a sustainable basis [2].

To examine the grievances that people shared on X (formerly X(formerly Twitter)) as urban grievances, Isha et al. suggested a framework based on NLP, which allowed the authorities to identify civic issues that were raised during online conversations [8].

Vanisree and Veerraghava explored the use of AI-driven communication technology in enhancing the interaction between citizens and local governments by using social media [15].

Multimodal and advanced artificial intelligence has been investigated as well to enhance civic complaint intelligence systems in recent studies. Khan et al. have designed Civic Dock, a multilingual system that facilitates voice, text, and image-powered submission of complaints to enable them to be properly redirected and traced [3].

A zero-shot large language model framework that can classify grievances based on their multimodality, as well as score their urgency, without significant training data was proposed by Rajkumar et al. [13].

Breja explored AI-based security and privacy systems to handle multimedia and multimodal data in the e-governance systems [4].

Prathyush et al. proposed a platform, called CivicEye, a combination of AI and blockchain to provide a transparent and tamper-proof system to report and resolve civic issues [12].

III. PROPOSED SYSTEM

The proposed Smart City Concern system functions as a comprehensive, multimodal civic grievance intelligence platform that automates the handling of complaints, including their processing, prioritization, and routing to the appropriate departments. In contrast to traditional grievance mechanisms that depend largely on manual review, this design incorporates Natural Language Processing (NLP), transformer-based zero-shot classification, sentiment analysis, Optical Character Recognition (OCR), and social media mining within a single integrated workflow.

The system gets the complaints through the two main channels (i) an online complaint portal and (ii) X(formerly Twitter) (X) posts mentioning or referring to the municipal authorities. The platform helps track civic concerns real time by incorporating the submission of structured forms with unstructured content on social media..

A. Multimodal Complaint Representation

Every case of complaint is represented as:

$$C = \{T, I, M\} \quad (1)$$

where:

- T denotes written expression of the complaint,
- I denotes optional image evidence,
- M denotes metadata including the time, user ID and location.

Such integrity means that the system can handle both text and visual cues at the same time.

Automated department classification involves automated systems that identify and classify departments to enable quick processing of the information.

B. Automated Department Classification

Automated Department Classification Automated department classification comprises automated systems that recognise and divide the departments to facilitate a fast processing of the information. A transformer-based model of zero-shot classification is employed in order to automatically redirect a complaint to the right municipal departments. Given a text of complaints, T , and a fixed label set, $L = \{l_1, l_2, \dots, l_k\}$, representing the civic departments (Sanitation, Roads, Water Supply, Electricity, and Public Safety), the classifier assigns predictions in the following way:

$$y = \text{ZeroShot}(T, L) \quad (2)$$

Output probability distribution is calculated by the use of softmax:

$$P(y|T) = \text{softmax}(Wh + b) \quad (3)$$

This method allows expansion of categories flexibly without retraining the model, so it can be used in changing smart city infrastructures.

C. Named Entity Recognition for Structured Extraction

To convert unstructured complaints into actionable tickets, Named Entity Recognition (NER) is applied:

$$E = \text{NER}(T) \quad (4)$$

The system extracts:

- Location references
- Infrastructure keywords
- Department indicators
- Urgency-related terms

Extracted entities are stored as structured fields in the database, improving traceability and analytics.

D. Sentiment-Based Priority Scoring

Citizen complaints often express frustration, urgency, or distress. Sentiment polarity is therefore used to estimate priority levels.

$$S = \text{Sentiment}(T) \quad (5)$$

Priority is determined as:

$$\text{Priority} = \begin{cases} \square & \text{High, } S < -0.6 \\ \square\square & \text{Medium, } -0.6 \leq S < 0 \\ \square\square\square & \text{Low, } S \geq 0 \end{cases} \quad (6)$$

This ensures emotionally critical complaints receive faster administrative attention.

E. OCR-Based Image Text Extraction

When image evidence I is provided, OCR is applied to extract embedded textual information:

$$T_I = \text{OCR}(I) \quad (7)$$

The extracted text is merged with complaint text:

$$T' = T + T_I \quad (8)$$

This enables the system to interpret information from signboards, notices, or infrastructure markings that may not be explicitly described in the complaint text.

F. Duplicate Complaint Detection

To reduce redundancy and administrative overload, duplicate detection is performed using TF-IDF vectorization:

$$V = \text{TFIDF}(T') \quad (9)$$

Similarity between complaints i and j is computed as:

$$\text{Sim}(i, j) = \frac{V_i \cdot V_j}{\|V_i\| \|V_j\|} \quad (10)$$

If similarity exceeds a threshold $\tau = 0.85$, the complaint is marked as a duplicate and linked to an existing ticket.

G. X(formerly Twitter) Integration Module

The system continuously monitors civic-related posts via X(formerly Twitter) API using predefined keywords and municipal mentions:

Extracted tweets undergo the same processing pipeline—classification, NER, sentiment scoring, and duplication check—ensuring parity between web-based and social media grievances.

Each layer operates independently yet communicates through RESTful APIs, ensuring modular scalability and deployment flexibility.

IV. METHODOLOGY

The Smart City Concern system (Fig. 1.) is developed as a smart system of civic grievances management, which transforms uncoded citizen complaints into formalized and operational service requests. The architecture is layered whereby the components carry out a particular role within the complaint processing pipeline. This modularity enhances scalability, maintenance and effective deployment within smart city environments.

Figure 1 below (system architecture) shows the general process of the proposed system. It starts with the so-called Citizen Interaction Layer, in which complaints are placed via a web portal or via social media (such as X(formerly Twitter)) (X). Users are able to leave a textual description of civic issues and may also leave pictures accompanying this to be used as supporting evidence.

The gathered grievances are then passed to Data Acquisition and Standardization Layer, which transforms all the inputs into harmonized representation to be analyzed. All the complaints can be expressed as:

$$C = \{T, I, M\} \quad (11)$$

The entry will be defined as follows: In case an image is included, the Optical Character Recognition (OCR) is used to retrieve text:

$$T_I = \text{OCR}(I) \quad (12)$$

The text extracted is mixed with the text of the complaint to generate an enriched representation:

$$T' = T + T_I \quad (13)$$

The standardized complaint is further handled by the Intelligent Processing Layer. In this case, it can be demonstrated that the complaint is analyzed through Natural Language Processing methods. A zero-shot classification model with the help of transformers automatically classifies the complaint to the corresponding municipal department:

$$D = \text{fclassify}(T') \quad (14)$$

Moreover Named Entity Recognition identifies the relevant entities like place names and infrastructure names and sentiment analysis identifies the level of urgency of the complaint.

In order to avoid duplication of records, the Similarity and Validation Layer detects duplicates by TF-IDF vectorization and Cosine similarity:

$$V = \text{TFIDF}(T') \quad (15)$$

$$\text{Sim}(i, j) = \frac{V_i \cdot V_j}{\|V_i\| \|V_j\|} \quad (16)$$

When the similarity score is greater than a preset threshold, the complaint is connected to an existing ticket instead of making a duplicate entry.

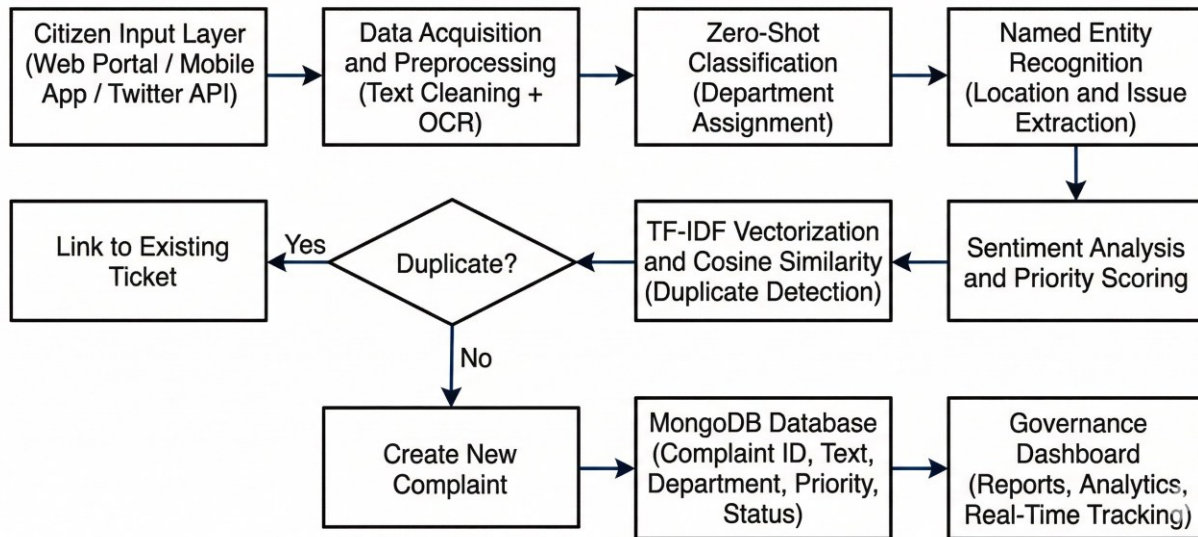


Fig. 1. Architecture of the Smart City Concern System.

The received and processed complaints are logged in the **Data Persistence Layer** that is based on MongoDB to provide the structured storage of complaint contents in the form of complaint ID, assigned department, extracted entities, level of priority, and resolution status.

Lastly, the Governance and Monitoring Interface Layer offers an administrative dashboard to the municipal authorities. The dashboard shows the analytics of complaints, division-by-department, filters of priorities, and a track of status, allowing the authorities to track the civic issues effectively and make adequate decisions.

Altogether, the suggested architecture will provide the multimodal data processing, automated classification of complaints, and real-time monitoring of social media into one AI-based system, enhancing the control of grievance in cities in terms of transparency and efficiency.

V. IMPLEMENTATION

The Smart City Concern system was implemented as a modular AI-driven grievance intelligence platform integrating web technologies, transformer-based NLP models, social media APIs, and database services. The development was structured to ensure scalability, reproducibility, and real-time responsiveness.

A. Technology Stack

The system was implemented using the following components:

- **Frontend:** Web interface developed using Streamlit for rapid prototyping and interactive visualization.
- **Backend:** Python-based services built with Flask to handle API routing and request processing.
- **Database:** MongoDB used for structured complaint storage and metadata indexing.

- **NLP Models:** HuggingFace Transformers library for zero-shot classification and sentiment analysis.
- **OCR Engine:** Tesseract OCR integrated for extracting textual information from complaint images.
- **Deployment Environment:** Ubuntu 22.04 with optional GPU acceleration for transformer inference.

The modular separation between frontend, processing engine, and storage layer allows independent scaling of components when complaint volume increases.

B. Dataset Description

The data set that was used to measure the Smart City Concern system was produced based on a mixture of publicly accessible civic complaint datasets and artificial generated complaints that represented typical urban problems including sanitation, road damage, water leakage, electricity fault, and safety issue in the community. X(formerly Twitter) posts that included the reference to municipal authorities were also gathered and combined with the web portal data on complaints to create a blend of structured submissions and unstructured posts on social media.

The data was separated into training, validation and testing dataset in the proportions of 70: 15: 15. All the complaints were sorted into five civic departments, which included Sanitation, roads and infrastructure, water supply, electricity and public safety to assist in model training and evaluation. The dataset was created in such a way that it represented all the categories equally in order to have a fair performance measure.

Figure 2 shows the distribution of complaints across different civic departments. Sanitation and Roads categories contained the highest number of complaints due to their frequent occurrence in urban environments. This classification structure enables the proposed AI system to automatically route complaints to the appropriate municipal department.

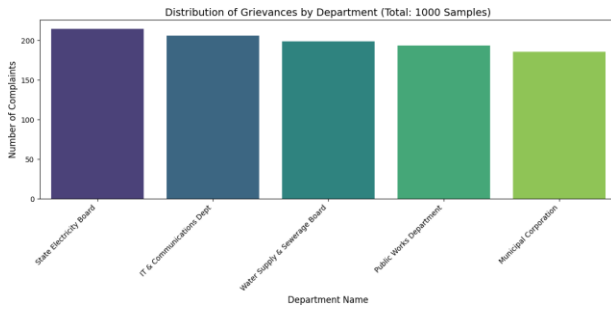


Fig. 2. Distribution of Complaints Across Municipal Departments

C. Text Preprocessing Pipeline

Before AI processing, incoming complaints were normalized through:

- Lowercasing and whitespace normalization
- Removal of non-informative special characters
- URL and hashtag filtering (for X(formerly Twitter) inputs)
- Stopword removal (optional during similarity detection)

If an image attachment was detected, OCR processing extracted embedded text and appended it to the complaint body for unified analysis.

D. Zero-Shot Department Classification

Zero-shot classification was done using a pre-trained transformer model (facebook/bart-large-mnli). Instead of training a department-specific classifier to perform this task, candidate classes were offered as a pre-defined set of labels of departments, which expanded dynamically without the need to retrain the model.

E. Named Entity Recognition

The NER pipeline based on transformer was used to extract:

- Location names
- Infrastructure references
- Service-related keywords

Entities extracted were then placed in the database as structured attributes, as JSON, to support analytics and filtering.

F. Sentiment-Based Priority Assignment

A sentiment-classifier that is based on DistilBERT was employed to calculate the polarity scores between -1 and +1. Priority thresholds were defined as:

- High Priority: $S < -0.6$
- Medium Priority: $-0.6 \leq S < 0$
- Low Priority: $S \geq 0$

This prioritization was automatic and therefore greatly decreased manual urgency assessment.

G. Duplicate Detection Module

The TF-IDF vectorization and cosine similarity scoring were used to identify duplicate complaints. In case similarity was above the threshold, a new complaint was not created and was associated with an existing ticket.

H. X(formerly Twitter) Integration

X(formerly Twitter) API v2 was used to track civic-related posts and hashtags. Retrieved tweets also received the same treatment as the web submissions.

The rate limiting and batching controls were introduced, which guaranteed API compliance and consistent ingestion.

I. Database Design

MongoDB collections were structured as follows:

- Complaint ID
- Cleaned Complaint Text
- Extracted Entities
- Assigned Department
- Priority Level
- Duplicate Reference (if any)
- Status Flag (Open / In Progress / Resolved)
- Timestamp Metadata

To achieve the maximum performance in dashboard filtering, indexes were made on department and priority fields.

VI. RESULTS AND DISCUSSION

Smart City Concern system was tested on the dataset of civic complaints based on simulated website portal entries and posts on X(formerly Twitter). The evaluation focused on the performance of different machine learning and deep learning models for complaint classification. The main objective was to determine which model provides the most accurate department routing for civic grievances.

TABLE I
MODEL COMPARISON FOR COMPLAINT CLASSIFICATION

Model	Accuracy
Naïve Bayes	72%
Support Vector Machine (SVM)	79%
Random Forest	84%
DistilBERT Classifier	89%
Zero-Shot Transformer (Proposed)	92.8%

Table I shows the comparison of different models used for civic complaint classification. Traditional machine learning models such as Naïve Bayes and SVM achieved moderate accuracy. Ensemble methods such as Random Forest performed better due to improved feature learning. However, transformer-based models demonstrated superior contextual understanding.

Figure 3 shows the comparison of accuracy achieved by different models. The graph clearly illustrates that transformer-based approaches outperform traditional machine learning models in civic complaint classification tasks.

The figure 4 determines the precision, recall, and F1-score of various classification models. Naive Bayes and the SVM are traditional models with moderate performance are characterized by poor contextual knowledge. Random Forest is better at improving results by ensuring that it captures more complex patterns. Deep learning systems like DistilBERT work more effectively due to the ability to grasp semantic context. The best result is the proposed Zero-Shot Transformer

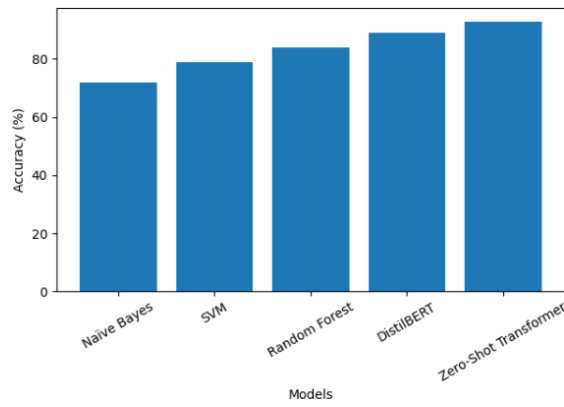


Fig. 3. Accuracy Comparison of Different Classification Models

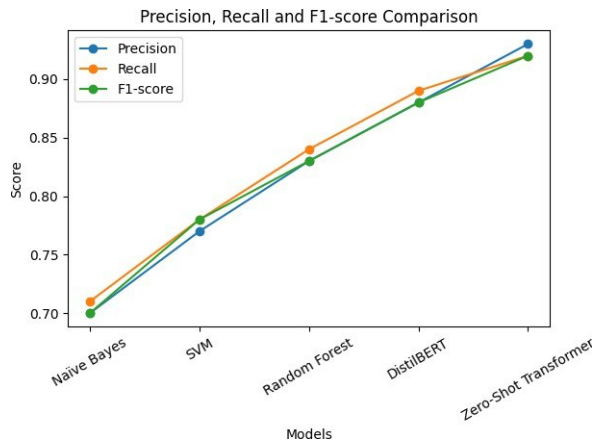


Fig. 4. Precision, Recall and F1-score Comparison Across Models

that effectively identifies the contextual meaning, as well as the ability to correlate complaints and department labels. Due to their ease of use and effectiveness in text-based tasks, traditional machine learning models like Naïve Bayes and Support Vector Machines (SVM) were first assessed for civic complaint classification. However, these models' capacity to capture contextual meaning in complaint descriptions is limited by their heavy reliance on manually engineered features and bag- of-words representations. By merging several decision trees to identify more intricate relationships in the data, Random Forest enhances performance. Even with this advancement, conventional models still have trouble deciphering colloquial language, misspellings, and brief complaint texts that are frequently present in social media posts and citizen submissions.

Deep learning-based language models, like DistilBERT, were investigated for complaint classification in order to overcome these constraints. Contextual embeddings and transformer mechanisms are used by BERT-based architectures to help the model comprehend the semantic relationships between words in a sentence. In natural language tasks involving unstructured and informal text, this capability greatly enhances classification performance. By enabling classification without

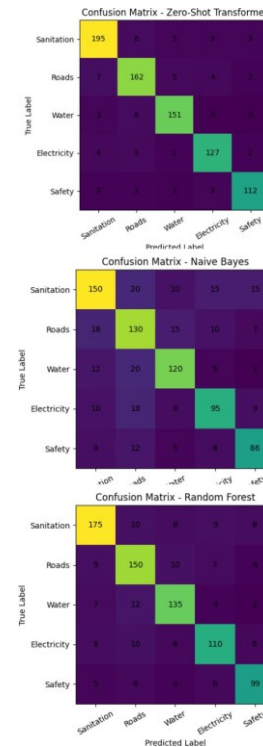


Fig. 5. Confusion Matrix for Model Performance

the need for task-specific retraining, the suggested Zero-Shot Transformer model expands on this strategy and increases flexibility. The system can dynamically match complaint descriptions with municipal department labels by utilising pre-trained language representations, which improves accuracy and increases the dependability of civic grievance routing. The confusion matrices demonstrate the classification abilities of various models in the categories of civic complaints. The machine learning models of earlier times like the Naive Bayes have more misclassification rates since they do not have the understanding of the context in the text of complaints. Random Forest is better at the accuracy of prediction since it delegates more intricate feature associations. Nevertheless, the suggested Zero-Shot Transformer model exhibits the most significant diagonal dominance which means the accuracy of classification and ability to better understand the description of civic complaints.

The proposed Zero-Shot Transformer model achieved the highest accuracy of 92.8%, showing its ability to understand complaint context and classify issues correctly without task-specific retraining.

A. Sentiment-Based Priority Evaluation

In addition to classification, sentiment analysis was used to determine the urgency level of complaints. Complaints containing strong negative sentiment or expressions of frustration were automatically marked as high priority.

Azhar Khan
@khan_Congress12

@GHMOnline @MC_Cyberabad

During ramazan no sweeping & no garbage lifting near masjid e asha
Firdous shatam ka talab Mohammedia lines Tolichowki Golconda
request to make it lift



Fig. 6. Sample Input Complaint from Web Portal or X(formerly Twitter)

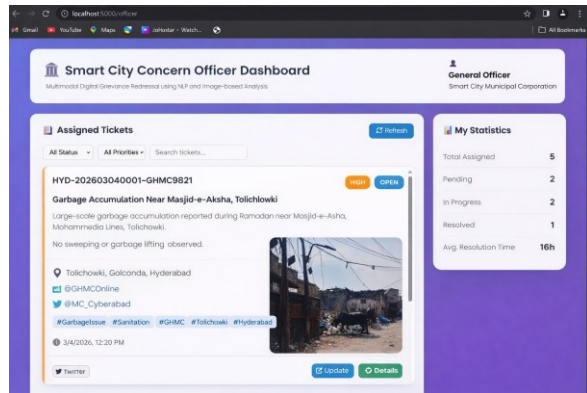


Fig. 7. Structured Complaint Output Generated by the System

B. Duplicate Complaint Detection

TF-IDF vectorization and cosine similarity were assessed as tools of detecting a duplicate complaint. The system was around 89 percent accurate in recognizing similar complaints. Majority of the similar complaints were noted in frequently reported problems like potholes, accumulation of garbage and power outages. The system saves a lot of work on the part of administration as it connects similar complaints to one ticket.

Upon processing every complaint, the system creates a structured ticket of complaint that consists of information extracted and classification results.

Figure 6 shows an example complaint submitted by a citizen through the web interface or collected from X (formerly Twitter). Figure 7 shows the structured output generated by the Smart City Concern system. The output contains the cleaned complaint text, assigned department, extracted entities, and priority level.

Overall, the research results demonstrate that the proposed Smart City Concern system significantly improves civic

grievance management by automating complaint classification, prioritization, and duplicate detection while maintaining high accuracy and efficiency.

VII. LIMITATIONS

The Smart City Concern system has also some limitations despite its good performance. Informal language, sarcasm or missing context in a social media complaint may influence model accuracy and the performance of OCR depends on image quality. Moreover, a textual similarity and sentiment-based prioritization-based duplicate detection approach can not necessarily be able to identify semantic similarity or actual urgency, and large-scale application will need substantial computational resources and consistent API connectivity.

VIII. CONCLUSION AND FUTURE SCOPE

This research work has proposed the Smart City Concern, a civic grievance intelligence system composed of multiple modes, where Artificial Intelligence (AI) is used to combine web-based complaint filing with X (formerly Twitter) surveillance. The suggested framework is a combination of transformer-based zero-shot classification, Named Entity Recognition, sentiment-oriented prioritization, OCR-based image processing, and duplicate detection that allow the automatic processing and redirecting of civic complaints to the respective municipal departments. It was experimentally proved that the system is characterized by a high level of classification accuracy and a significant decrease in the manual load and allows the creation of complaints almost in real time. The layered architecture supports modular deployment, scalability, and integrative efficiency into smart city governance structures. To make the system efficient in the future, it should be facilitated to handle multilingual processing of complaints so as to handle regional languages to enhance citizen involvement. Deep embedding-based similarity models can also enhance the accuracy of duplicate complaints detection even more than the traditional TF-IDF. The geo-spatial analytics and visualization of a heatmap might be used to assist authorities in identifying the areas at risk within the city and the repetitive infrastructure problems. Moreover, large metropolitan settings can be scaled by implementing lightweight transformer styles and cloud processing platforms. It could also be addressed in future employment with automated response generation and smart escalation processes to further simplify the process of managing civic grievances.

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